

INTRODUCTION TO ROBOTICS

Title: Homogeneous Transformation in Planar and Spatial Mechanisms using Python Robotics Toolbox

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Aim

To understand and implement homogeneous transformations for planar (2D) and spatial (3D) mechanisms using the Python Robotics Toolbox. The experiment demonstrates translation, rotation, combined transformations and visualization of coordinate frames using homogeneous transformation matrices.

Learning Objectives

- Understand homogeneous transformation matrices.
- Perform translation in planar and spatial coordinate systems.
- Perform rotation in planar and spatial coordinate systems.
- Combine translation and rotation into a single transformation.
- Compare different orders of transformation.
- Visualize transformed coordinate frames.

Software Requirements

- Python 3.10 or later
- Robotics Toolbox for Python
- SpatialMath Python Package
- NumPy
- Matplotlib
- Jupyter Notebook / VS Code

Theory

A robot manipulator moves in space by changing its position and orientation.

Instead of handling rotation and translation separately, robotics combines both into a **Homogeneous Transformation Matrix (HTM)**.

The homogeneous transformation matrix is represented as

$$T = \begin{bmatrix} R & P \\ 0 & 1 \end{bmatrix}$$

where

- **R** = Rotation Matrix
- **P** = Position Vector

The Robotics Toolbox provides

- **SE2** → Planar Transformations (2D)
- **SE3** → Spatial Transformations (3D)

Algorithm

1. Task 1: Create planar translation matrices for the given coordinate values.
2. Task 2: Generate planar rotation matrices for different angles.
3. Task 3: Combine translation followed by rotation using SE2.
4. Task 4: Compare Translation×Rotation and Rotation×Translation.
5. Task 5: Create 3D translation matrices using SE3.
6. Task 6: Perform rotations about the X-axis.
7. Task 7: Perform rotations about the Y-axis.
8. Task 8: Perform rotations about the Z-axis.
9. Task 9: Combine translation with X, Y and Z rotations.
10. Task 10: Visualize transformed coordinate frames.
11. Challenge: Apply multiple transformations, obtain the final HTM, determine end-effector position and verify the effect of changing transformation order.

Python Program

```
# Import required libraries
from spatialmath import SE2, SE3

# =====
# TASK 1 - TRANSLATION
# =====

# Different translations
translations = [
    (2,3),
    (5,4),
    (3,-2),
    (-4,6)
]

for i,(x,y) in enumerate(translations, start=1):
    print("\n-----")
    print(f"Trial {i}")
    print(f"Translation = ({x},{y})")
    T = SE2(x,y)
    print(T)
```

```
# =====
# TASK 2 - ROTATION
# =====

angles = [30,60,90,180]
for angle in angles:
    print("\n-----")
    print(f"Rotation = {angle} Degrees")
    R = SE2.Rot(angle,"deg")
    print(R)
```

```
# =====
# TASK 3 - TRANSLATION FOLLOWED BY ROTATION
# =====

cases = [
    ((3,1),45),
    ((5,2),60),
    ((2,6),120)
]
for (x,y),angle in cases:
    print("\n-----")
    print(f"Translation : ({x},{y})")
    print(f"Rotation      : {angle} Degrees")
    T = SE2(x,y)
    R = SE2.Rot(angle,"deg")
    H = T * R
    print(H)
```

```
# =====
# TASK 4 - ROTATION FOLLOWED BY TRANSLATION
# =====
T = SE2(4,2)
R = SE2.Rot([90,"deg"])
print("\nTranslation then Rotation")
print(T * R)
print("\nRotation then Translation")
print(R * T)
```

```
# =====
# TASK 5 - TRANSLATION IN 3D
# =====
points = [
    (4,2,1),
    (5,5,5),
    (-2,3,7)
]
for x,y,z in points:
    print("\n-----")
    print(f"Translation ({x},{y},{z})")
    T = SE3(x,y,z)
    print(T)
```

```
# =====
# TASK 6 - ROTATION ABOUT X AXIS
# =====

angles = [30,90,180]
for angle in angles:
    print("\n-----")
    print(f"Rotation about X = {angle}")
    Rx = SE3.Rx(angle,"deg")
    print(Rx)
```

```
# =====
# TASK 7 - ROTATION ABOUT Y AXIS
# =====
angles = [45,120,180]
for angle in angles:
    print("\n-----")
    print(f"Rotation about Y = {angle}")
    Ry = SE3.Ry(angle,"deg")
    print(Ry)
```

```
# =====
# TASK 8 - ROTATION ABOUT Z AXIS
# =====

angles = [60,90,270]
for angle in angles:
    print("\n-----")
    print(f"Rotation about Z = {angle}")
    Rz = SE3.Rz(angle,"deg")
    print(Rz)
```

```
# =====
# TASK 9 - COMBINED TRANSFORMATION
# =====
examples = [
    ((5,4,2),(30,45,60)),
    ((2,8,3),(45,30,90)),
    ((1,5,4),(60,60,60))
]
for (x,y,z),(ax,ay,az) in examples:
    print("\n-----")
    print(f"Translation : ({x},{y},{z})")
    print(f"Rx={ax} Ry={ay} Rz={az}")
    T = SE3(x,y,z)
    Rx = SE3.Rx(ax,"deg")
    Ry = SE3.Ry(ay,"deg")
    Rz = SE3.Rz(az,"deg")
    H = T * Rx * Ry * Rz
    print(H)
```

```
# =====
# TASK 10 - COORDINATE FRAME VISUALIZATION
# =====

T = SE3(2,3,4) * SE3.Rz(45,"deg")
print("Plotting Coordinate Frame...")
T.plot(frame='A')
```

```

# =====
# CHALLENGE EXERCISE
# =====

# First Translation
T1 = SE3(3,2,1)

# Rotations
Rx = SE3.Rx(45,"deg")
Ry = SE3.Ry(60,"deg")
Rz = SE3.Rz(30,"deg")

# Second Translation
T2 = SE3(1,2,0)

# Final Homogeneous Transformation
Final = T1 * Rx * Ry * Rz * T2

print("\nFinal Homogeneous Transformation Matrix\n")
print(Final)

# Extract final position
print("\nFinal Position of End Effector")
print(Final.t)

# Plot
print("\nDisplaying Final Coordinate Frame...")

Final.plot(frame='Final')

# Compare order
Changed = T2 * Rz * Ry * Rx * T1

print("\nTransformation after changing order\n")
print(Changed)

```

Outputs

- Task 1 – Translation matrices (4 trials)

```

-----
Trial 1
Translation = (2,3)
  1      0      2
  0      1      3
  0      0      1

-----

Trial 2
Translation = (5,4)
  1      0      5
  0      1      4
  0      0      1

-----

Trial 3
Translation = (3,-2)
  1      0      3
  0      1     -2
  0      0      1

-----

Trial 4
Translation = (-4,6)
  1      0     -4
  0      1      6
  0      0      1

```

- Task 2 – Rotation matrices (30°, 60°, 90°, 180°)

```

-----
Rotation = 30 Degrees
  0.866  -0.5    0
  0.5    0.866  0
  0      0      1

-----

Rotation = 60 Degrees
  0.5    -0.866  0
  0.866  0.5    0
  0      0      1

-----

Rotation = 90 Degrees
  0      -1      0
  1      0      0
  0      0      1

-----

Rotation = 180 Degrees
 -1      0      0
  0      -1     0
  0      0      1

```

- Task 3 – Combined Translation → Rotation matrices

```

-----
Translation : (3,1)
Rotation   : 45 Degrees
  0.7071   -0.7071   3
  0.7071    0.7071   1
  0         0        1

-----

Translation : (5,2)
Rotation   : 60 Degrees
  0.5       -0.866    5
  0.866     0.5       2
  0         0        1

-----

Translation : (2,6)
Rotation   : 120 Degrees
 -0.5       -0.866    2
  0.866     -0.5      6
  0         0        1

```

- Task 4 – Comparison of $T \times R$ and $R \times T$

```

Translation then Rotation
  0      -1      4
  1       0      2
  0       0      1

Rotation then Translation
  0      -1     -2
  1       0      4
  0       0      1

```

- Task 5 – 3D Translation matrices

```

-----
Translation (4,2,1)
  1      0      0      4
  0      1      0      2
  0      0      1      1
  0      0      0      1

-----

Translation (5,5,5)
  1      0      0      5
  0      1      0      5
  0      0      1      5
  0      0      0      1

-----

Translation (-2,3,7)
  1      0      0     -2
  0      1      0      3
  0      0      1      7
  0      0      0      1

```


- Task 6 – Rotation about X-axis

Rotation about X = 30			
1	0	0	0
0	0.866	-0.5	0
0	0.5	0.866	0
0	0	0	1

Rotation about X = 90			
1	0	0	0
0	0	-1	0
0	1	0	0
0	0	0	1

Rotation about X = 180			
1	0	0	0
0	-1	0	0
0	0	-1	0
0	0	0	1

- Task 7 – Rotation about Y-axis

Rotation about Y = 45			
0.7071	0	0.7071	0
0	1	0	0
-0.7071	0	0.7071	0
0	0	0	1

Rotation about Y = 120			
-0.5	0	0.866	0
0	1	0	0
-0.866	0	-0.5	0
0	0	0	1

Rotation about Y = 180			
-1	0	0	0
0	1	0	0
0	0	-1	0
0	0	0	1

- Task 8 – Rotation about Z-axis

Rotation about Z = 60			
0.5	-0.866	0	0
0.866	0.5	0	0
0	0	1	0
0	0	0	1

Rotation about Z = 90			
0	-1	0	0
1	0	0	0
0	0	1	0
0	0	0	1

Rotation about Z = 270			
0	1	0	0
-1	0	0	0
0	0	1	0
0	0	0	1

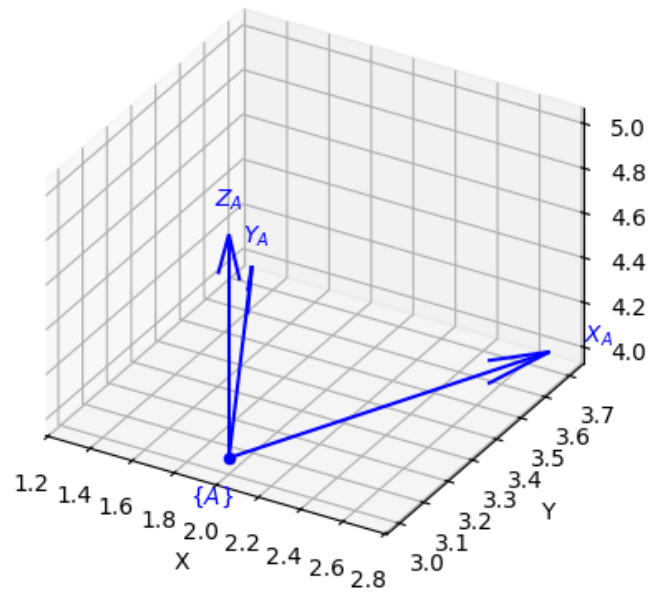
- Task 9 - Combined transformation matrices

Translation : (5,4,2)			
Rx=30 Ry=45 Rz=60			
0.3536	-0.6124	0.7071	5
0.9268	0.1268	-0.3536	4
0.1268	0.7803	0.6124	2
0	0	0	1

Translation : (2,8,3)			
Rx=45 Ry=30 Rz=90			
0	-0.866	0.5	2
0.7071	-0.3536	-0.6124	8
0.7071	0.3536	0.6124	3
0	0	0	1

Translation : (1,5,4)			
Rx=60 Ry=60 Rz=60			
0.25	-0.433	0.866	1
0.808	-0.3995	-0.433	5
0.5335	0.808	0.25	4
0	0	0	1

- Task 10 – Coordinate frame plots (A, B and C)



- Challenge Exercise – Final transformation matrix, end-effector position and final coordinate frame plot

Final Homogeneous Transformation Matrix

0.433	-0.25	0.866	2.933
0.8839	0.3062	-0.3536	3.496
-0.1768	0.9186	0.3536	2.66
0	0	0	1

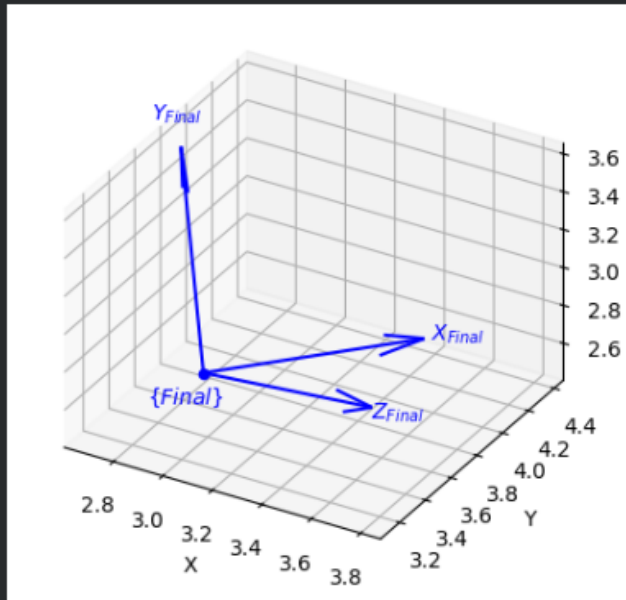
Final Position of End Effector

[2.9330127 3.49625591 2.66034061]

Displaying Final Coordinate Frame...

Transformation after changing order

0.433	0.1768	0.8839	3.536
0.25	0.9186	-0.3062	4.281
-0.866	0.3536	0.3536	-1.537
0	0	0	1



Observations

- Translation changes only the position of the frame.
- Rotation changes only the orientation of the frame.
- Homogeneous transformation combines translation and rotation in a single matrix.
- The order of multiplication affects the final result because matrix multiplication is not commutative.
- SE3 enables representation of rigid body motion in three-dimensional space.
- Coordinate frame visualization helps verify the correctness of transformations.

Result

The experiment was successfully completed. Planar and spatial homogeneous transformations were implemented using the SpatialMath package. Translation, rotation, combined transformations and coordinate-frame visualization were successfully performed. It was observed that changing the order of transformations changes the final transformation matrix.

Answers to Task 4 Questions

1. Are the two matrices identical?

No. Translation followed by rotation and rotation followed by translation produce different matrices.

2. Which coordinate changed?

The position coordinates change because the reference frame changes.

3. Why does the order matter?

Transformation multiplication is not commutative; each transformation is applied relative to the current coordinate frame.